Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. NIST SP 800-30 Rev. 1 is used to guide the risk analysis of the information system.

Purpose

Consider the following questions to help you write:

- *How is the database server valuable to the business?*
- *Why is it important for the business to secure the data on the server?*
- *How might the server impact the business if it were disabled?*

This remote database server stores valuable business data that many employees access remotely from around the world. Employees regularly query data from the server to help them find potential customers. The database has been open to the public for the last 3 years since it launched. It is important for this database to be secured because it holds valuable information that can be leaked and used against the business, also to prevent unauthorized access. If the server was to be disabled somehow then operations that rely on the server would be disrupted, customers could be lost, and there could be significant financial loss.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Perform reconnaissance and surveillance of organization	*3*	*2*	*6*
Malicious Software	Install malicious software to capture or exfiltrate data from the database server.	*3*	*3*	*9*
Network or Infrastructure Outage	Disrupt access to the database server due to a network or infrastructure outage.	*2*	*2*	*4*

Approach

Considering that the remote database server is publicly accessible and contains valuable business data, threats such as hackers, malicious software, and network outages pose significant risks to the organization. Because the database has been public for three years, attackers could be gathering information and performing reconnaissance for a future attack. Malicious software could exploit public access to capture sensitive data, while network outages could disrupt remote employees from accessing the server. These threats could significantly impact business operations and assets, and the public nature of the database increases the likelihood of exploitation.

Remediation Strategy

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using strong passwords, role-based access controls, and multi-factor authentication to limit user privileges. Encryption of data in motion using TLS instead of SSL. IP allow-listing to corporate offices to prevent random users from the internet from connecting to the database. To mitigate network or infrastructure outages, the organization should implement regular backups of the database server so if one fails another can take over and data is not lost.